

Why is the body cavity of a cockroach not a true coelom but a haemocoel?

Fat bodies

The haemocoel contains many large sized fat bodies. These bodies are also called **corpora adiposa**. These are similar to the liver of the vertebrates in certain functions. A fat body has many lobules containing four types of cells namely **trophocytes** (store food) **mycetocytes** (contain symbiotic bacteria), **oenocytes** (synthesise and store lipids), **urate cells** (store uric acid).

7.3 LOCOMOTION

Periplaneta has double mode of locomotion. 1-Running (cursorial locomotion) 2-Flying.

1. Running or cursorial locomotion

Cockroach is a cursorial insect and it can run on the ground with the help of legs. During walking or running cockroach forms two **tripods** with the legs. Each tripod is formed by foreleg and hind leg of one side and middle leg of the other side. During locomotion, the three legs of one tripod are kept on the ground and the other three legs of another tripod are carried forward. In this process, the foreleg and the hind leg of the tripod pull and push the body while the middle leg acts as a **pivot**. Thus cockroach moves using the two tripods alternately. The animal does not slip back as plantulae, pulvilli and claws provide grip. The claws and the arolium help in locomotion on **rough surfaces** whereas plantulae are useful on **smooth surfaces**.

How is each tripod of cockroach formed during locomotion?

2. Flying: Though cockroach is a cursorial insect, it can fly over short distances with the help of its wings. While flying the first pair of wings is stretched out at right angles to the body. The second pair of wings is moved up and down during flight. Wings are elevated by the contraction of dorsoventral muscles. Contraction of the dorso-longitudinal muscles and relaxation of the dorsoventral muscles depresses the wings.

7.4 DIGESTIVE SYSTEM OF PERIPLANETA

The digestive system of cockroach consists of an alimentary canal and the associated glands. The preoral cavity, surrounded by the mouth parts, is present in front of the mouth. The hypopharynx divides it into two chambers called **cibarium** (anterior) and **salivarium** (posterior).

I. Alimentary canal

The alimentary canal of cockroach is a long tube and is coiled at some places. It extends between the mouth and the anus. It is divided into three regions, namely, foregut

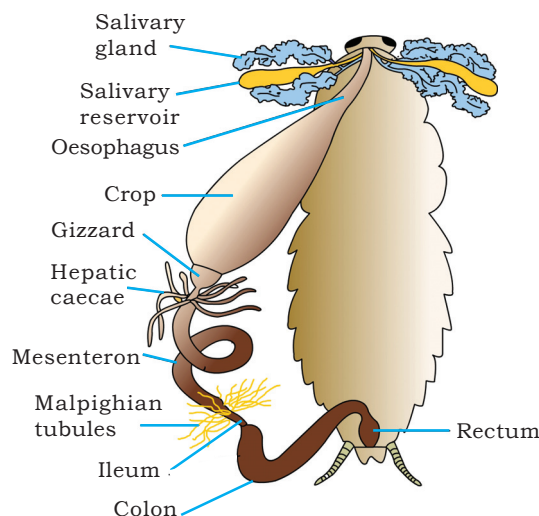


Fig. 7.4 *Periplaneta* : Digestive system

or **stomodaeum**, midgut or **mesenteron** and hindgut or **proctodaeum**. The foregut and hindgut are internally lined by ectoderm. The mesenteron is lined by the endodermal cells.

1. Foregut or stomodaeum

The foregut includes pharynx, oesophagus, crop, and gizzard. It is internally lined by a chitinous cuticle. Mouth opens into the **pharynx**, which in turn leads into a narrow tubular oesophagus. The **oesophagus** opens behind into a thin walled distensible sac called **crop**. The crop serves as a reservoir for storing food. Its outer surface is covered by a network of tracheae.

Behind the crop there is a thick walled muscular **proventriculus**, or **gizzard**. The chitinous inner lining of the gizzard has six powerful teeth, which form an efficient grinding apparatus. Behind each tooth is a hairy pad, which bears backwardly directed bristles. Among these plates, food is thoroughly ground into fine particles. These food particles are filtered by the bristles. The gizzard thus acts both as a **grinding mill** and also as a **sieve**. There is a membranous projection of the gizzard into the mesenteron in the form of a funnel called **stomodeal valve**. This valve prevents the entry (regurgitation) of food from the mesenteron back into the gizzard.

2. Midgut (mesenteron or ventriculus)

The midgut is a short and narrow tube behind the gizzard. It is also called **mesenteron** or **ventriculus**. Between the ventriculus and the gizzard, arising from ventriculus, there are six to eight finger like diverticula called **hepatic caecae**. They are helpful in digestion and absorption of the digested food materials. Ventriculus is functionally divided into an anterior

The secretory part of the ventriculus has many gland cells and it secretes several enzymes. The 'bolus' of food in the mesenteron is enveloped by a chitinous and porous membrane called **peritrophic membrane**, which is secreted by the mid gut. The peritrophic membrane is a network of chitin fibrils in a glycoprotein matrix, secreted by cells in the anterior region of the midgut.

Digested food is absorbed into the blood through the peritrophic membrane in the posterior absorptive region of the ventriculus. The peritrophic membrane protects the wall of the ventriculus from hard food particles in the food. The opening of the ventriculus into the hindgut is controlled by a **sphincter muscle**. It prevents entry (regurgitation) of undigested food and uric acid from the hindgut into the midgut.

Which part of the alimentary canal produces the peritrophic membrane around the food bolus in *Periplaneta*?

3. Hindgut or proctodaeum

The hindgut is a long coiled tube, consisting of three regions namely **ileum**, **colon** and **rectum**. It is internally lined by chitinous cuticle. The ileum that lies behind the mesenteron is a short tube. Six bundles of fine yellow, blind tubules called **Malpighian tubules** open into the ileum near the junction of mesenteron and ileum. Malpighian tubules are excretory in function. Ileum collects uric acid from the malpighian tubules and undigested food from the mesenteron. Ileum opens behind into a long coiled tube called colon. Colon leads into a short and wide rectum, which opens out through the anus. Rectum bears on its inner side six longitudinal folds called rectal papillae. They are concerned with the reabsorption of water from the undigested food.

II. Digestive glands

The digestive glands associated with the alimentary canal of cockroach are salivary glands, hepatic caecae and glandular cells of the mesenteron.

1. Salivary glands

There is a pair of salivary glands attached to the ventrolateral sides of the crop, one on each side. Each salivary gland has two lobes. Each lobe of salivary gland has many lobules called **acini**. Each acinus is a group of secretory cells called **zymogen cells** with a small ductule. The ductules of both the lobes of a salivary gland unite to form a **common salivary duct** on each side.

The two common salivary ducts are joined to form the **median salivary duct**. Between the two lobes of a salivary gland of each side is a sac called **salivary receptacle** that stores saliva. It leads into a **receptacular duct**, or 'reservoir duct'. The receptacular ducts of both the sides are united to form a **common receptacular duct**, or 'common reservoir duct'. The median salivary duct opens into the common receptacular duct. Later these two form an **efferent salivary duct**. The efferent salivary duct opens at the base of the hypopharynx. Acinar cells secrete saliva, which contains starch digesting enzymes such as **amylase**.

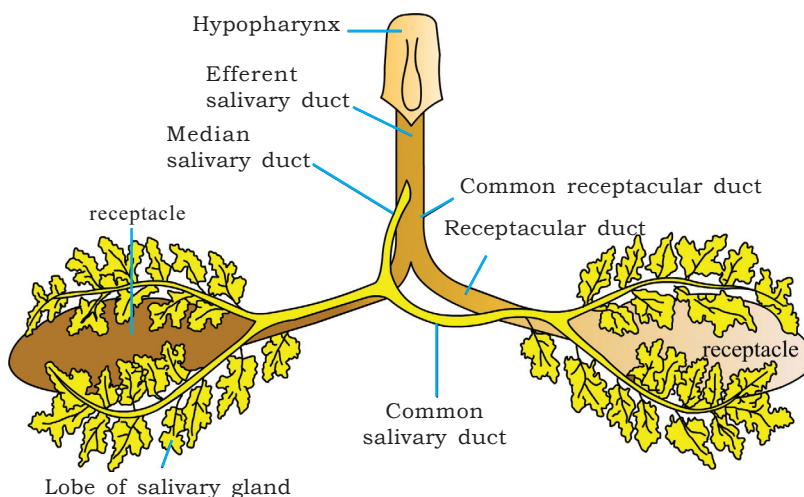


Fig. 7.5 *Periplaneta* : Salivary glands